

Experimental Design

The experiment has been pre-registered on the 4th of May 2025 through OSF (registration number: X4zuk), after collecting participation from schools but prior to the activities. The experiment has been approved by the Bioethics Committee of the University of Turin on the 15th of April 2025 (protocol n. 0336761).

The experiment was structured as a randomized controlled trial (RCT) with three treatment arms.

Four primary schools located in the Metropolitan City of Turin were invited to participate through an email sent to school principals. The four schools contacted were not randomly selected from the overall population but represent a convenience sample based on availability of direct contact with the principals. The population from which the sample was drawn therefore consists of all children aged 8 to 10 years enrolled in those primary schools. Participation was conditional on voluntary acceptance by the school and the class teachers.

Out of the contacted schools, three agreed to participate, for a total of twelve classes across cohort four and five. An additional class of the third cohort was included for a separate focus group aimed at qualitatively exploring the behavioral mechanisms. The focus group discussions were not part of the experimental evaluation but served as qualitative triangulation; notes were taken during the sessions.

Randomization was clustered at the class level with the sample being stratified by school and cohort, so that classes of the same school and cohort were randomly assigned to one of the three experimental conditions: Game treatment, Lecture treatment, or Pure control. When only two classes from the same school participated, random assignment was restricted to the first two conditions. The treatment was clustered at the class level to account by design for potential differences among schools; it was not randomized at the individual level to avoid spillovers between same class members.

The experiment was conducted by the same two researchers in all sessions, who alternated roles as Experiment Facilitators across treatments to ensure procedural consistency. Participation was voluntary, and all classes were self-selected into the study following the invitation email. Importantly, this applies to all treatment arms, including the control group. This design minimizes potential selection bias due to teachers' motivation or willingness to participate, as within each school, the decision to join the experiment was made collectively by the same teaching staff. It was also common knowledge for the teaching staff that all classes would be able to participate to the game and the lecture at some point in time. We agreed with the teachers that the control group would receive the treatments, both the game and the lecture, only after the final survey. This allowed us to maintain the interest in participation also from the control group.

Participants and teachers were informed that the activity was part of a study on environmental education. However, the specific objective of comparing different teaching methods was not disclosed ex ante to prevent expectancy effects. Children were informed of their assigned activity only at the beginning of the session.

The activities were carried out during the same day for each school involved in the experiment. Teachers were asked to arrange the daily class activity so that the interaction between different classes from the moment of the first survey and the moment of the final survey was minimized (e.g. breaks were taken at different times for each class involved in the experiment). This was done to reduce potential spillovers between groups on the first day of activity. Yet, the experiment has been designed over two days of activity, taking place two weeks one after the other.

The assumption that only the treated are affected (SUTVA) does not hold, but potential spillovers would have a positive effect, therefore the estimated treatment effect would be downward bias. Still, we account for the spillovers in the econometric analysis.

Treatment 1: Game

In the first treatment, participants played the educational board game H2gO!. Each session lasted approximately 30 minutes of game play, with around 15 minutes before and after to prepare the materials and the tables where to dispose the game. The two researchers acted as Experiment Facilitators, introducing the rules and assisting players during the game. Each class (around 20 students) was divided into two groups, and students played in groups of two or three, that acted as single players. Each group cooperated with another group, allowing the participation of 16–24 children simultaneously. The groups were created endogenously by the children, so that they could play with their friends. While this does not follow a random allocation, it better replicates would happen in reality if children were to play autonomously. When the children were not able to decide in short time whit whom to play, we formed the group based on how they were sitting in the class - keep kids that were sitting close to each other in the same group.

During the gameplay, participants were instructed to read their cards aloud to ensure uniform access to information.

Treatment 2: Lecture

The second treatment consisted of a 30-minute lecture on water conservation, focusing on the interconnection between climate, technology, and water availability. The lecture was

interactive: children were periodically asked to answer short questions to maintain engagement. A PowerPoint presentation was used to support the lecture, replicating the same definitions and concepts presented in the game to ensure content equivalence across treatments. Fonts and colors were chosen to be dyslexia- and color-blind-friendly, following the same accessibility design as the game materials. The lecture also made use of some materials, such as a glass of water, to make some examples with the help of some physical materials to which the children could rely and recall more easily. In particular, the glass of water was used to explain the concept that of the overall water in the globe, only 1% is freshwater that is available for use to drink and use - therefore 1 glass over 100. Moreover, the glass of water was also used to explain what happens to the stock of freshwater available depending on the weather and the technologies that we use: when there is a sunny day, the water in the glass diminishes, while when there is a flooding it increases so much that the glass cannot contain it all; with accumulation technologies we switch the glass with a box that can hold more water when it rains, while with efficiency technologies we keep more water in the glass by using it better.

The lecture was held by the same two researchers who acted as game facilitators in the Treatment 1.

Treatment 3: Control

The third arm served as a pure control group and did not receive any treatment at the time of the first data collection. Control classes completed the same surveys as the treated groups but were exposed to the educational activities only after the two-weeks follow up survey, to ensure ethical fairness and maintain interest in participation. At the time of accepting to participate in the experiment, all teachers were aware that their classes would have received both treatments regardless of the treatment arm they would end up in, to allow for self-selection also of the control group.